

An Introduction to Wireless Multimedia Sensor Networks

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Tiny wireless devices and the enormous growth of wireless communication technologies have already established the stage for large-scale deployment of wireless sensor networks (WSNs). A typical WSN consists of a large number of small, low-cost sensor nodes, which are distributed in the target area for collecting data of interest. Most of the time, WSN is used for monitoring, tracking and event management related applications. WSN is not a new topic as many inventions have been done and countless applications have been successfully implemented.

Fig. 1 and Table I show the similarities and differences between ad hoc wireless networks, wireless sensor networks and wireless multimedia sensor networks (WMSNs). All these networks are used for small-scale and medium-scale networks that do not require a fixed architecture and can be deployed temporarily. WMSN requires a novel architecture, and the advanced protocol design and algorithms need to be implemented to fulfil the criteria. This article exposes the internal structure of the multimedia sensor node,

architecture of WMSN, their challenges, research opportunities and applications.

Internal structure of the multimedia sensor node

Normally, a multimedia sensor device includes a sensing unit, processing unit (CPU), communication module, co-ordination subsystem, storage unit and an optional mobility/actuation unit. Sensing units usually have two sub-units: sensors [digital micro cameras, microphones and/or scalar sensors (for example, temperature sensors, humidity sensors, fire sensors, pressure sensors and gas sensors)], and analogue to digital converters (ADCs). Most of the sensors generate analogue signals which should be converted into digital form by using ADCs as these are to be fed into the CPU.

The CPU executes the system software in charge of coordinating sensing and communication tasks for signal processing, and it is interfaced with a storage unit. A communication Tx/Rx module interfaces the device to the network. In many applications, it has been observed that around 70 per cent of the battery is consumed for transceiver processes and the remaining

30 per cent consumption includes computational tasks, signal processing, co-ordination systems and sensor sub-units.

TABLE I
Comparison between Ad Hoc Wireless Networks, WSNs and WMSNs

	Ad hoc wireless networks	WSNs	WMSNs
Node size	Large	Very small	Small
Node density	Very low	High	Medium
Node coverage area	Medium (metres)	Low (metres)	Medium (metres)
Data type	Huge data, like files and audio-visual information	Scalar, like temperature, humidity and fire sensors, among others	Multimedia, like still image, video streaming and voice
Bandwidth requirement	High	Low	Medium
Congestion control algorithm requirement	High	Low	High
Power consumption	Does not matter	Very low	Low

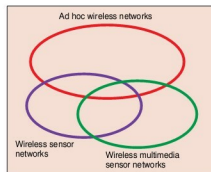


Fig. 1: Classification of different types of wireless networks